

UNIT 5

INTRODUCTION TO SECURITY POLICIES AND CYBER LAWS : Need for An Information Security Policy, Introduction to Indian Cyber Law, Objective and Scope of the Digital Personal Data Protection Act 2023, Intellectual Property Issues, Overview of Intellectual Property Related

1 NEED FOR AN INFORMATION SECURITY POLICY

1.1 What is an Information Security Policy?

An Information Security Policy (ISP) is a formal written document that defines:

- What information needs protection
- Who is responsible for protecting it
- How protection should be done
- What happens if policy is violated

Simple meaning: It is the **security rulebook of an organization**.

1.2 Why is Information Security Policy Needed?

Today organizations store:

- Customer data
- Financial data
- Employee records
- Business secrets
- Research data

Without policy:

- Employees may share passwords
- Data may be copied to personal devices
- No standard security procedure
- Legal penalties for data breach

Example:

If a company does not define password rules:

Employees may use password like

"company123" → Easy to hack → Data breach → Legal penalty under DPDP Act 2023

1.3 Objectives of Security Policy

1. Protect Confidentiality (no unauthorized access)
2. Maintain Integrity (data not altered)
3. Ensure Availability (system accessible)
4. Define accountability
5. Meet legal compliance

1.4 Types of Security Policies

1. Organizational Policy

High-level rules for entire organization.

2. Issue-specific Policy

Focused on specific issue. Example:

- Email usage policy
- Social media policy

3. System-specific Policy

Technical configuration rules. Example:

- Firewall rules
- Server configuration

1.5 Important Components

- Password policy
- Data classification policy
- Backup and recovery policy
- Incident response policy
- Access control policy
- Remote work policy

Engineering Connection:

When building backend system: Security policy ensures:

- Password hashing
- Role-based access control
- Logging and monitoring
- Data encryption

Security must be policy-driven, not random.

2 INTRODUCTION TO INDIAN CYBER LAW

Main cyber law in India:

✦ **Information Technology Act 2000**

2.1 Why IT Act Was Introduced?

Before 2000:

- No law for online crimes
- No legal status for digital signatures
- No recognition of e-contracts

IT Act 2000 was introduced to:

- Provide legal recognition to electronic records
- Punish cyber offenses
- Promote e-commerce

2.2 Important Sections (Conceptual Understanding)

- Section 43: Unauthorized access
- Section 66: Hacking offenses
- Section 66C: Identity theft
- Section 66D: Cheating by impersonation
- Section 67: Publishing obscene content

2.3 IT Act 2008 Amendment

Added:

- Cyber terrorism
- Data protection concepts
- Stronger penalties

Example:

If someone hacks college website → Punishable under IT Act.

3 DIGITAL PERSONAL DATA PROTECTION ACT 2023

✦ **Digital Personal Data Protection Act 2023**

Very important modern law.

3.1 Why This Act Was Needed?

Due to:

- Massive data collection
- Social media platforms
- Online payments
- Data leaks

Earlier: No strong personal data protection law in India.

3.2 Objective of DPDP Act 2023

- Protect digital personal data
- Regulate data processing
- Give rights to individuals
- Impose penalties for misuse

3.3 Important Definitions

Data Principal

Person whose data is collected.

Example: You using Instagram.

Data Fiduciary

Company collecting data.

Example: Instagram company.

3.4 Rights of Data Principal

1. Right to know how data is used
2. Right to correction
3. Right to erase data
4. Right to grievance redressal

3.5 Obligations of Company

- Obtain consent
- Protect data
- Report data breaches
- Appoint data protection officer (if required)

3.6 Penalties

Heavy fines (up to crores).

Example:

If fintech company leaks Aadhaar + bank data → Heavy penalty under DPDP Act.

4 INTELLECTUAL PROPERTY ISSUES

4.1 What is Intellectual Property?

Intellectual Property (IP) is legal protection for creations of mind.

Examples:

- Software code
- Research algorithm
- Mobile app
- Logo
- Design

4.2 Why IP Important in Cyber World?

Because:

- Digital content easy to copy
- Software piracy common
- Online copyright violations frequent

Example:

If someone copies your startup app code and uploads on Play Store, it is IP violation.

5 OVERVIEW OF IP LEGISLATION IN INDIA

Important Acts:

1. Indian Patents Act 1970
2. Copyright Act 1957
3. Trade Marks Act 1999

6 PATENT

Governed by: 🚩 Indian Patents Act 1970

6.1 What is Patent?

Patent gives exclusive rights for 20 years to inventor.

6.2 What Can Be Patented?

- New technology
- Industrial process
- Innovative hardware

Software patents in India are limited unless combined with hardware.

Example:

New cybersecurity encryption hardware chip → patent possible.

COPYRIGHT

Governed by:  **Copyright Act 1957**

7.1 What is Protected?

- Software source code
- Website content
- Books
- Music
- Films

Duration: Author's lifetime + 60 years.

Example:

If someone copies your GitHub project without permission → copyright violation.

TRADEMARK

Governed by:  **Trade Marks Act 1999**

8.1 What is Trademark?

Trademark protects:

- Brand name
- Logo
- Symbol

- Slogan

Example:

If someone creates fake Amazon website with similar logo → Trademark violation + phishing crime.

9 INTELLECTUAL PROPERTY ISSUES IN CYBERCRIME

1. Software piracy
2. Torrent movie downloads
3. Logo misuse
4. Domain name spoofing
5. Code plagiarism

Good. Now I'll give you **most important difference tables from Unit-5 (Security Policies & Cyber Laws)**.

These are **very high probability exam questions (5–10 marks)**. Write at least 5–6 points in exam.

1 Security Policy vs Cyber Law

Basis	Security Policy	Cyber Law
Definition	Internal rules of organization	Government legal framework
Authority	Organization management	Parliament / Government
Scope	Limited to organization	Applies to whole country
Enforcement	Internal disciplinary action	Legal punishment
Example	Password policy	IT Act 2000

② IT Act 2000 vs Digital Personal Data Protection Act 2023

Basis	IT Act 2000	DPDP Act 2023
Purpose	Regulate cybercrime & e-commerce	Protect personal digital data
Focus	Cyber offenses	Data privacy
Scope	All cyber activities	Personal data processing
Penalties	For hacking, fraud etc.	For data breaches
Nature	Broad cyber law	Data protection law

③ Data Principal vs Data Fiduciary

Basis	Data Principal	Data Fiduciary
Meaning	Person whose data is collected	Organization collecting data
Role	Data owner	Data controller
Rights	Access, correction, erase	Must protect data
Example	Instagram user	Instagram company

④ Patent vs Copyright vs Trademark

Basis	Patent	Copyright	Trademark
Protects	Invention	Creative work	Brand identity
Duration	20 years	Life + 60 years	10 years (renewable)
Example	New encryption device	Software code	Company logo
Registration	Mandatory	Automatic (creation)	Mandatory

Governing Law	Patents Act 1970	Copyright Act 1957	Trade Marks Act 1999
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🔥 Very Important Table – Almost always asked.

5 Copyright vs Patent

Basis	Copyright	Patent
Protects	Expression of idea	Idea/invention
Registration	Not compulsory	Compulsory
Duration	Life + 60 years	20 years
Example	App code	New hardware invention
Cost	Low	Expensive

6 Trademark vs Copyright

Basis	Trademark	Copyright
Protects	Brand name/logo	Creative content
Validity	Renewable	Fixed duration
Example	Brand logo	Website content
Purpose	Prevent brand misuse	Prevent copying

7 Organizational Policy vs Issue-Specific Policy

Basis	Organizational Policy	Issue-Specific Policy
Scope	Entire organization	Specific issue
Example	Overall security rules	Email usage policy

Level	High-level	Detailed rules
Coverage	Broad	Narrow

8 Confidentiality vs Privacy

Basis	Confidentiality	Privacy
Meaning	Protect data from unauthorized access	Right of individual to control data
Focus	Security control	Personal rights
Example	Encryption	Consent before data use
Law Related	Security policy	DPDP Act

9 Civil Offense vs Criminal Offense (Cyber Context)

Basis	Civil Offense	Criminal Offense
Nature	Compensation based	Punishment based
Example	Data negligence	Hacking
Outcome	Fine	Jail + Fine
Law Section	IT Act civil sections	IT Act criminal sections

10 Personal Data vs Sensitive Personal Data

Basis	Personal Data	Sensitive Personal Data
Meaning	Basic personal information	Highly confidential info

Example	Name, email	Biometric, health data
Risk Level	Moderate	High
Protection Level	Required	Very strict

1 1 Digital Signature vs Electronic Signature

Basis	Digital Signature	Electronic Signature
Technology	Cryptography based	Any electronic method
Security	High	Moderate
Legal Recognition	Under IT Act	Also recognized
Example	DSC token	OTP based consent